Chapter 3 — Indefinite Integration: Detailed Teaching Notes

Maharashtra HSC · Mathematics & Statistics Part II (Arts & Science) · Textbook pp. 95–150 · File 1 of 3

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0. How to Use These Notes

Scope. This file covers the *entire* Chapter 3 of the Std XII Part II textbook (pages 95–150): definition and elementary formulas (Ex 3.1), integration by substitution (Ex 3.2 A), special integrals and quadratic forms (Ex 3.2 B), linear-over-quadratic forms (Ex 3.2 C), integration by parts (Ex 3.3), partial fractions (Ex 3.4), the generalised by-parts rule (§3.5) and the Miscellaneous Exercise. Definite integration (Chapter 4) is *not* included.

Notation (same as the textbook). $\log x$ always means the natural logarithm $\log_e x$. c is the constant of integration. I write $\log|f(x)|$ with modulus bars; the textbook often drops them (e.g. it writes $\log(\sec x + \tan x)$). Both are accepted in the board exam — the modulus is simply the mathematically complete form.

Study order. Read each section's *Intuition* box first, then the formula/theorem, then attempt each worked example yourself *before* reading the solution. Every red **Common Mistake** box is a place where students actually lose marks.

0.1 What the board actually asks from this chapter

I checked all six board papers in your project folder. Every Indefinite Integration question is listed below — these are **verified PYQs**, read directly from the papers.

Paper (code)	1 mark	2 marks	3 marks	4 marks
Mar 2024 (J-862)	$\int \frac{dx}{x^2+25}$	$\int \log x\,dx$	Prove $\int \frac{dx}{a^2-x^2}$	$\int \frac{5e^x\,dx}{(e^x+1)(e^{2x}+9)}$
Jul 2024 (J-174)	$\int \frac{5^x}{3^x}dx$	$\int \frac{e^x(1+x)}{\sin^2(xe^x)}dx$	$\int \frac{\sin(x+a)}{\cos(x-b)}dx$	$\int x^2 \sin 3x\,dx$
Feb 2025 (J-312)	—	MCQ $\int x^x(1+\log x)dx$; $\int x^9 \sec^2(x^{10})dx$; $\int \frac{dx}{25-9x^2}$	—	$\int e^{\sin^{-1}x}\frac{x+\sqrt{1-x^2}}{\sqrt{1-x^2}}dx$
Jul 2025 (J-384)	$\int \frac{dx}{\sqrt{x}}$	$\int \frac{dx}{x+x^{-10}}$	$\int \frac{\sin(x+a)}{\cos(x-b)}dx$	Prove $\int \sqrt{a^2-x^2}\,dx$ formula
Feb 2026 (J-165)	$\int \frac{2x}{1+x^2}dx$	$\int \sqrt{1+\sin 2x}\,dx$	$\int \frac{\sin x}{\sin 3x}dx$	$\int \frac{3x^2+4x-5}{(x^2-1)(x+2)}dx$
Jun 2026 (J-276)	—	MCQ $\int f\cdot g$ with $e^{\sin^{-1}x}$; $\int x\log x\,dx$; $\int \frac{e^x(1+x)}{\cos^2(xe^x)}dx$	—	$\int \frac{2\,dx}{3+2\sin^2 x+5\cos^2 x}$

What this pattern tells you (observation, not official weightage)

- In every one of the six papers, about **10 marks** of questions come from this chapter, usually one each of 1, 2, 3 and 4 marks. Sections B, C and D have internal choice, so you can choose, but these are easy marks to bank.
- **Repeats are real:** $\int \frac{\sin(x+a)}{\cos(x-b)}dx$ appeared in *Jul 2024 and Jul 2025*; $\int \frac{e^x(1+x)}{\sin^2(xe^x)}$ and its $\cos^2$ twin appeared in *Jul 2024 and Jun 2026*; $\int \frac{3x^2+4x-5}{(x^2-1)(x+2)}dx$ (Feb 2026) is a *textbook solved example*;

$\int \frac{dx}{3+2\sin^2 x+5\cos^2 x}$ (Jun 2026) is also a textbook solved example.

- 4-mark questions come from: partial fractions, integration by parts, the $\frac{1}{a\sin^2 x+b\cos^2 x+c}$ type, the $e^x[f+f']$ type, and **proofs of standard formulas**.
- This does not guarantee anything about the next paper. It tells you where the examiners' habits lie.

1. Prerequisites You Must Have Ready

Integration is differentiation run backwards. If your derivatives are shaky, integration will be shaky. Revise these before continuing.

1.1 Derivatives you will reverse

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
x^n	nx^{n-1}	$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
e^x	e^x	$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
a^x	$a^x \log a$	$\tan^{-1} x$	$\frac{1}{1+x^2}$
$\log x$	$\frac{1}{x}$	$\cot^{-1} x$	$-\frac{1}{1+x^2}$
$\sin x$	$\cos x$	$\sec^{-1} x$	$\frac{1}{x\sqrt{x^2-1}}$
$\cos x$	$-\sin x$	$\operatorname{cosec}^{-1} x$	$-\frac{1}{x\sqrt{x^2-1}}$
$\tan x$	$\sec^2 x$	$\sec x$	$\sec x \tan x$
$\cot x$	$-\operatorname{cosec}^2 x$	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

1.2 Trigonometric identities used in this chapter

- $\sin^2 x+\cos^2 x=1$, $\quad 1+\tan^2 x=\sec^2 x$, $\quad 1+\cot^2 x=\operatorname{cosec}^2 x$
- $\sin 2x=2\sin x\cos x$, $\quad \cos 2x=\cos^2 x-\sin^2 x=1-2\sin^2 x=2\cos^2 x-1$
- **Power reduction:** $\sin^2 x=\dfrac{1-\cos 2x}{2}$, $\quad \cos^2 x=\dfrac{1+\cos 2x}{2}$
- **Triple angle:** $\sin 3x=3\sin x-4\sin^3 x$, $\quad \cos 3x=4\cos^3 x-3\cos x$
- **Half-angle forms:** $1+\cos x=2\cos^2\frac{x}{2}$, $\quad 1-\cos x=2\sin^2\frac{x}{2}$, $\quad 1\pm\sin x=\left(\cos\frac{x}{2}\pm\sin\frac{x}{2}\right)^2$
- **Product to sum:** $2\sin A\cos B=\sin(A+B)+\sin(A-B)$, $2\cos A\cos B=\cos(A+B)+\cos(A-B)$, $2\sin A\sin B=\cos(A-B)-\cos(A+B)$
- **Compound angle:** $\sin(A\pm B)=\sin A\cos B\pm\cos A\sin B$, $\cos(A\pm B)=\cos A\cos B\mp\sin A\sin B$
- $\tan\frac{x}{2}$ **substitution:** $\sin x=\dfrac{2t}{1+t^2}$, $\cos x=\dfrac{1-t^2}{1+t^2}$ where $t=\tan\frac{x}{2}$

1.3 Log and exponent rules

$\log(mn)=\log m+\log n$, $\log\frac{m}{n}=\log m-\log n$, $\log m^k=k\log m$, $e^{\log f(x)}=f(x)$, $a^{\log_a f(x)}=f(x)$.

2. What Integration Means

Intuition — the reverse gear

Differentiation answers: *"Given a function, what is its rate of change?"* Integration answers the reverse question: *"Given the rate of change, what was the original function?"*
Analogy: a speedometer shows your speed at every instant (the derivative). If someone hands you the full speedometer record, you can reconstruct how far you travelled (the original function) — **except** you cannot know where you *started*. That unknown starting point is exactly the constant c.

Definition (Textbook §3.1)

If $\frac{d}{dx}\,g(x) = f(x)$, then $g(x)$ is called a **primitive** (antiderivative) of $f(x)$, and we write

$$\int f(x)\,dx = g(x) + c$$

read as *"the integral of $f(x)$ with respect to x is $g(x) + c$"*. Here $f(x)$ is the **integrand**, $g(x)$ the **integral**, and c an **arbitrary constant**.

Why the $+c$? Since $\frac{d}{dx}(\text{constant}) = 0$, the functions x^2, x^2+4 and x^2-5 all have derivative $2x$. So $\int 2x\,dx$ is not one function but a whole *family* $x^2 + c$ — the curves are vertical shifts of each other, and at any fixed x (say $x = 2$) all of them have the same tangent slope $2(2) = 4$. (This is Fig. 3.1.1 in the textbook.)

The self-check that never fails

Differentiate your answer. If you get back the integrand, your answer is right. Do this mentally for every integration you write in the exam; it takes ten seconds and catches sign errors and missing factors like $\frac{1}{a}$.

Common Mistake

Forgetting $+c$ in an indefinite integral. Examiners can deduct marks for it. Write $+c$ on *every* final answer of an indefinite integral.

3. Elementary Formulas and Rules (Ex 3.1)

3.1 The standard table

Each formula below comes from reversing a derivative (Textbook §3.1.1). The right column is the **linear extension**: replace x by $(ax+b)$ and divide by a.

Integrand	$\int f(x)\,dx$	$\int f(ax+b)\,dx$
$x^n\ (n \neq -1)$	$\dfrac{x^{n+1}}{n+1} + c$	$\dfrac{(ax+b)^{n+1}}{(n+1)\,a} + c$
$\dfrac{1}{x}$	$\log\lvert x\rvert + c$	$\dfrac{1}{a}\log\lvert ax+b\rvert + c$
e^x	$e^x + c$	$\dfrac{1}{a}e^{ax+b} + c$
$A^x\ (A > 0)$	$\dfrac{A^x}{\log A} + c$	$\dfrac{A^{ax+b}}{a\log A} + c$
$\cos x$	$\sin x + c$	$\dfrac{1}{a}\sin(ax+b) + c$
$\sin x$	$-\cos x + c$	$-\dfrac{1}{a}\cos(ax+b) + c$
$\sec^2 x$	$\tan x + c$	$\dfrac{1}{a}\tan(ax+b) + c$
$\operatorname{cosec}^2 x$	$-\cot x + c$	$-\dfrac{1}{a}\cot(ax+b) + c$

Integrand	$\int f(x)\,dx$	$\int f(ax+b)\,dx$
$\sec x \tan x$	$\sec x + c$	$\frac{1}{a}\sec(ax+b)+c$
$\operatorname{cosec} x \cot x$	$-\operatorname{cosec} x + c$	$-\frac{1}{a}\operatorname{cosec}(ax+b)+c$
$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x + c$	—
$\frac{1}{1+x^2}$	$\tan^{-1} x + c$	—
$\frac{1}{x\sqrt{x^2-1}}$	$\sec^{-1} x + c$	—

Signs of the trig integrals — the "co-answer gets a minus" rule

If the **answer** is a "co"-function (cos, cot, cosec), it carries a **minus** sign: $\int \sin x\,dx = -\cos x$, $\int \operatorname{cosec}^2 x\,dx = -\cot x$, $\int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x$. If the answer is sin, tan or sec, there is no minus. (Reason: the derivatives of cos, cot, cosec are the negative ones.)

3.2 The three linearity theorems (Textbook §3.1.2)

Theorems 1, 2, 3

If f and g are integrable and k is a constant:

1. $\int [f(x)+g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$
2. $\int [f(x)-g(x)]\,dx = \int f(x)\,dx - \int g(x)\,dx$
3. $\int k f(x)\,dx = k\int f(x)\,dx$

Proof of Theorem 1

Let $\int f(x)\,dx = g_1(x)+c_1$ and $\int g(x)\,dx = g_2(x)+c_2$. By definition,

$$\frac{d}{dx}[g_1(x)+c_1] = f(x), \qquad \frac{d}{dx}[g_2(x)+c_2] = g(x).$$

The derivative of a sum is the sum of the derivatives, so

$$\frac{d}{dx}\big[(g_1(x)+c_1)+(g_2(x)+c_2)\big] = f(x)+g(x).$$

By the definition of integration, $\int [f(x)+g(x)]\,dx = (g_1(x)+c_1)+(g_2(x)+c_2) = \int f(x)\,dx + \int g(x)\,dx$. Theorems 2 and 3 are proved in exactly the same way.

Common Mistake — there is NO product or quotient rule

$\int f(x)g(x)\,dx \neq \int f\,dx \cdot \int g\,dx$ and $\int \frac{f}{g}\,dx \neq \frac{\int f}{\int g}$. Products need **substitution** or **integration by parts**; quotients need algebra, substitution or **partial fractions**.

3.3 Technique catalogue for Ex 3.1 — "simplify first, then integrate"

Exercise 3.1 never needs a new method. Every question is a *disguise*: you rewrite the integrand into standard-table pieces. There are six disguises.

(a) Break into powers of x

Example 3.1 — $\int \left(\sin x + \frac{1}{x} + \frac{1}{\sqrt[3]{x}}\right) dx$ (Textbook)

Write $\dfrac{1}{\sqrt[3]{x}} = x^{-1/3}$, then use the power rule with $n = -\frac{1}{3}$:

$$\int \sin x\, dx + \int \frac{1}{x}\, dx + \int x^{-1/3} dx = -\cos x + \log|x| + \frac{x^{2/3}}{2/3} + c = -\cos x + \log|x| + \frac{3}{2}x^{2/3} + c.$$

(b) Use trig identities to reach a standard form

Example 3.2 — $\int (\tan x + \cot x)^2 dx$ (Textbook)

Expand: $(\tan x + \cot x)^2 = \tan^2 x + 2\tan x \cot x + \cot^2 x = \tan^2 x + 2 + \cot^2 x$ (because $\tan x \cot x = 1$).
Now replace $\tan^2 x = \sec^2 x - 1$ and $\cot^2 x = \operatorname{cosec}^2 x - 1$ (why: $\tan^2$ and $\cot^2$ are *not* in the table, but $\sec^2$ and cosec^2 are):

$$\int (\sec^2 x - 1 + 2 + \operatorname{cosec}^2 x - 1)\, dx = \int (\sec^2 x + \operatorname{cosec}^2 x)\, dx = \tan x - \cot x + c.$$

Example 3.3 — Odd powers: $\int \cos^3 x\, dx$ (Textbook)

From $\cos 3x = 4\cos^3 x - 3\cos x$ we get $\cos^3 x = \frac{1}{4}(\cos 3x + 3\cos x)$. Hence

$$\int \cos^3 x\, dx = \frac{1}{4}\left(\frac{\sin 3x}{3} + 3\sin x\right) + c = \frac{1}{12}\sin 3x + \frac{3}{4}\sin x + c.$$

(Similarly, $\sin^3 x = \frac{1}{4}(3\sin x - \sin 3x)$.)

Example 3.4 — Even powers: $\int \sin^4 x\, dx$ (Textbook)

Use power reduction twice. $\sin^4 x = \left(\frac{1-\cos 2x}{2}\right)^2 = \frac{1}{4}(1 - 2\cos 2x + \cos^2 2x)$, and $\cos^2 2x = \frac{1+\cos 4x}{2}$. So

$$\sin^4 x = \frac{1}{4}\left(\frac{3}{2} - 2\cos 2x + \frac{1}{2}\cos 4x\right).$$

$$\int \sin^4 x\, dx = \frac{1}{4}\left(\frac{3}{2}x - \sin 2x + \frac{1}{8}\sin 4x\right) + c = \frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + c.$$

Example 3.5 — Product to sum: $\int \sin 5x \cos 7x\, dx$ (Textbook)

$2\sin 5x \cos 7x = \sin 12x + \sin(-2x) = \sin 12x - \sin 2x$. So

$$\int \sin 5x \cos 7x\, dx = \frac{1}{2}\int (\sin 12x - \sin 2x)\, dx = -\frac{\cos 12x}{24} + \frac{\cos 2x}{4} + c.$$

Example 3.6 — Perfect-square trick: $\int \sqrt{1 + \sin 2x}\, dx$ [PYQ Feb 2026, 2 marks]

Write $1 = \sin^2 x + \cos^2 x$ and $\sin 2x = 2\sin x \cos x$:

$$1 + \sin 2x = \sin^2 x + \cos^2 x + 2\sin x \cos x = (\sin x + \cos x)^2.$$

Taking the square root (assuming $\sin x + \cos x \geq 0$ on the interval considered, as the textbook does),

$$\int \sqrt{1 + \sin 2x}\, dx = \int (\sin x + \cos x)\, dx = -\cos x + \sin x + c = \sin x - \cos x + c.$$

The textbook's version of this idea is $\int \sqrt{1+\sin 3x}\,dx$: there $1+\sin 3x = \left(\cos\frac{3x}{2}+\sin\frac{3x}{2}\right)^2$, giving $\frac{2}{3}\left(\sin\frac{3x}{2}-\cos\frac{3x}{2}\right)+c$. **Jul 2024 asked the definite version** $\int_0^{\pi/4}\sqrt{1+\sin 2x}\,dx$ (that is Chapter 4, but the first step is identical).

(c) Multiply by the conjugate

Example 3.7 — $\int \frac{1}{1-\sin x}\,dx$ (Textbook)

Multiply top and bottom by $(1+\sin x)$ so that the denominator becomes $1-\sin^2 x=\cos^2 x$:

$$\int \frac{1+\sin x}{\cos^2 x}\,dx = \int \left(\sec^2 x+\sec x\tan x\right)dx = \tan x+\sec x+c.$$

Same trick: $\int \frac{\cos x}{1-\cos x}dx$: multiply by $(1+\cos x)$, get $\int \frac{\cos x+\cos^2 x}{\sin^2 x}dx = \int(\operatorname{cosec} x\cot x+\operatorname{cosec}^2 x-1)\,dx = -\operatorname{cosec} x-\cot x-x+c$.

Example 3.8 — Rationalising: $\int \frac{1}{\sqrt{3x+1}-\sqrt{3x-5}}\,dx$ (Textbook)

Multiply by the conjugate $\sqrt{3x+1}+\sqrt{3x-5}$. The denominator becomes $(3x+1)-(3x-5)=6$:

$$I=\frac{1}{6}\int\left[(3x+1)^{1/2}+(3x-5)^{1/2}\right]dx=\frac{1}{6}\left[\frac{(3x+1)^{3/2}}{\frac{3}{2}\cdot 3}+\frac{(3x-5)^{3/2}}{\frac{3}{2}\cdot 3}\right]+c$$
$$=\frac{1}{27}\left[(3x+1)^{3/2}+(3x-5)^{3/2}\right]+c.$$

(The "$\cdot 3$" is the a of the linear extension.)

(d) Write the numerator in terms of the denominator

Example 3.9 — $\int \frac{2x+3}{5x-1}\,dx$ (Textbook)

Idea: make the numerator contain a copy of the denominator. $2x+3=\frac{2}{5}(5x-1)+\frac{2}{5}+3=\frac{2}{5}(5x-1)+\frac{17}{5}$. Then

$$\int\left[\frac{2}{5}+\frac{17/5}{5x-1}\right]dx=\frac{2}{5}x+\frac{17}{5}\cdot\frac{1}{5}\log|5x-1|+c=\frac{2}{5}x+\frac{17}{25}\log|5x-1|+c.$$

Example 3.10 — $\int \frac{2x-7}{\sqrt{3x-2}}\,dx$ (Textbook)

$2x-7=\frac{2}{3}(3x-2)-\frac{17}{3}$. So

$$I=\frac{2}{3}\int(3x-2)^{1/2}dx-\frac{17}{3}\int(3x-2)^{-1/2}dx=\frac{2}{3}\cdot\frac{(3x-2)^{3/2}}{\frac{3}{2}\cdot 3}-\frac{17}{3}\cdot\frac{(3x-2)^{1/2}}{\frac{1}{2}\cdot 3}+c$$
$$=\frac{4}{27}(3x-2)^{3/2}-\frac{34}{9}(3x-2)^{1/2}+c.$$

Example 3.11 — $\int \frac{x^3}{x-1}\,dx$ (Textbook)

Add and subtract 1 to create $x^3-1=(x-1)(x^2+x+1)$:

$$\int\frac{x^3-1+1}{x-1}dx=\int\left(x^2+x+1+\frac{1}{x-1}\right)dx=\frac{x^3}{3}+\frac{x^2}{2}+x+\log|x-1|+c.$$

(e) Exponential and log simplifications

Example 3.12 — $\int \frac{e^{4\log x}-e^{5\log x}}{x^5}\,dx$ and $\int \frac{5^x}{3^x}dx$

First: $e^{4\log x}=e^{\log x^4}=x^4$ and $e^{5\log x}=x^5$. So $\int \frac{x^4-x^5}{x^5}dx=\int\left(\frac{1}{x}-1\right)dx=\log|x|-x+c$.
Second [PYQ Jul 2024, 1 mark]: $\frac{5^x}{3^x}=\left(\frac{5}{3}\right)^x$, which is A^x with $A=\frac{5}{3}$:

$$\int\left(\frac{5}{3}\right)^x dx=\frac{(5/3)^x}{\log(5/3)}+c.$$

Similarly $\int \frac{3^x-4^x}{5^x}dx=\frac{(3/5)^x}{\log(3/5)}-\frac{(4/5)^x}{\log(4/5)}+c$ (Textbook).

(f) Inverse-trig simplifications

Example 3.13 — $\int \sin^{-1}(\cos 3x)\,dx$ (Textbook)

$\cos 3x=\sin\left(\frac{\pi}{2}-3x\right)$, so $\sin^{-1}(\cos 3x)=\frac{\pi}{2}-3x$ (taking the principal branch). Then

$$\int\left(\frac{\pi}{2}-3x\right)dx=\frac{\pi}{2}x-\frac{3x^2}{2}+c.$$

Example 3.14 — $\int \tan^{-1}\sqrt{\frac{1-\sin x}{1+\sin x}}\,dx$ (Textbook)

Write $\sin x=\cos\left(\frac{\pi}{2}-x\right)$ and use $1-\cos\theta=2\sin^2\frac{\theta}{2}$, $1+\cos\theta=2\cos^2\frac{\theta}{2}$:

$$\sqrt{\frac{1-\cos(\frac{\pi}{2}-x)}{1+\cos(\frac{\pi}{2}-x)}}=\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\ \Rightarrow\ \text{integrand}=\frac{\pi}{4}-\frac{x}{2}.$$

$$\int\left(\frac{\pi}{4}-\frac{x}{2}\right)dx=\frac{\pi}{4}x-\frac{x^2}{4}+c.$$

Example 3.15 — Find $f(x)$ if $f'(x)=x-\frac{3}{x^3}$ and $f(1)=\frac{11}{2}$ (Ex 3.1 IV)

$f(x)=\int\left(x-3x^{-3}\right)dx=\frac{x^2}{2}-3\cdot\frac{x^{-2}}{-2}+c=\frac{x^2}{2}+\frac{3}{2x^2}+c$.
Use the given value: $f(1)=\frac{1}{2}+\frac{3}{2}+c=2+c=\frac{11}{2}\Rightarrow c=\frac{7}{2}$. Hence $f(x)=\frac{x^2}{2}+\frac{3}{2x^2}+\frac{7}{2}$.
Lesson: a given value of f fixes the constant c. This is the only place in the chapter where c gets a number.

4. Integration by Substitution (Ex 3.2 A)

Intuition — spot the inner function and its derivative

Differentiation of a composite function uses the chain rule: $\frac{d}{dx}\sin(x^3)=\cos(x^3)\cdot 3x^2$. So if an integrand looks like **(function of an inner expression) × (derivative of that inner expression)**, it came from a chain rule, and we can undo it by renaming the inner expression as t.
Analogy: a gift box inside wrapping paper. Substitution peels off the wrapping (t = the inner function) so that what remains is a simple standard gift.

Theorem (§3.2.1)

If $x=\phi(t)$ is a differentiable function of t, then

$$\int f(x)\,dx=\int f[\phi(t)]\,\phi'(t)\,dt.$$

Proof

Since $x = \phi(t)$, $\dfrac{dx}{dt} = \phi'(t)$. Let $\int f(x)\,dx = g(x)$, so $\dfrac{d}{dx}[g(x)] = f(x)$. By the chain rule,

$$\frac{d}{dt}[g(x)] = \frac{d}{dx}[g(x)] \cdot \frac{dx}{dt} = f(x)\,\frac{dx}{dt} = f[\phi(t)]\,\phi'(t).$$

By the definition of integration, $g(x) = \int f[\phi(t)]\,\phi'(t)\,dt$, i.e. $\int f(x)\,dx = \int f[\phi(t)]\,\phi'(t)\,dt$.

4.1 The procedure (write these steps in the exam)

1. **Choose** t = the inner function whose derivative is (up to a constant) also present.
2. **Differentiate**: write $(\ldots)\,dx = dt$. Adjust constants, e.g. $3x^2dx = dt \Rightarrow x^2dx = \frac{1}{3}dt$.
3. **Replace** every x-expression by t. *No x may remain.*
4. **Integrate** in t.
5. **Resubstitute** t back in terms of x and add $+c$.

4.2 The four corollaries — memorise as patterns

Corollaries I–IV (§3.2.1)

Pattern	Result
I. $\int f(ax+b)\,dx$	$\frac{1}{a}\,g(ax+b)+c$, where $\int f(x)dx = g(x)$
II. $\int [f(x)]^n f'(x)\,dx$ $(n \neq -1)$	$\dfrac{[f(x)]^{n+1}}{n+1} + c$
III. $\int \dfrac{f'(x)}{f(x)}\,dx$	$\log\lvert f(x)\rvert + c$
IV. $\int \dfrac{f'(x)}{\sqrt{f(x)}}\,dx$	$2\sqrt{f(x)} + c$

Proofs of Corollaries II, III, IV (one substitution does all three)

Put $f(x) = t$, so $f'(x)\,dx = dt$.

- **II:** $\int [f(x)]^n f'(x)\,dx = \int t^n\,dt = \frac{t^{n+1}}{n+1} + c = \frac{[f(x)]^{n+1}}{n+1} + c$.
- **III:** $\int \frac{f'(x)}{f(x)}dx = \int \frac{dt}{t} = \log|t| + c = \log|f(x)| + c$.
- **IV:** $\int \frac{f'(x)}{\sqrt{f(x)}}dx = \int t^{-1/2}dt = 2\sqrt{t} + c = 2\sqrt{f(x)} + c$.

Corollary I: put $ax + b = t$, $a\,dx = dt$: $\int f(t)\frac{dt}{a} = \frac{1}{a}g(t) + c = \frac{1}{a}g(ax+b) + c$.

Corollary III in one line

"Top is the derivative of the bottom ⇒ answer is log of the bottom." Before using any other method on a fraction, *always* check whether the numerator is (a multiple of) the derivative of the denominator.

4.3 Worked examples

Example 4.1 — $\int \frac{2x}{1+x^2}\,dx$ [PYQ Feb 2026, 1 mark]

Numerator $2x$ is exactly the derivative of $1+x^2$. By Corollary III,

$$\int \frac{2x}{1+x^2}dx = \log(1+x^2)+c.$$

(No modulus needed since $1+x^2>0$.)

Example 4.2 — $\int x^9\sec^2(x^{10})\,dx$ [PYQ Feb 2025, 2 marks]

Inner function x^{10}; its derivative $10x^9$ is present except for the constant 10.
Put $x^{10}=t \;\Rightarrow\; 10x^9dx=dt \;\Rightarrow\; x^9dx=\frac{1}{10}dt$.

$$I=\frac{1}{10}\int \sec^2 t\,dt=\frac{1}{10}\tan t+c=\frac{1}{10}\tan(x^{10})+c.$$

Example 4.3 — $\int \frac{e^x(1+x)}{\sin^2(xe^x)}\,dx$ [PYQ Jul 2024] and the $\cos^2$ version [PYQ Jun 2026]

The inner function is xe^x. Its derivative (product rule) is $xe^x+e^x=e^x(1+x)$ — exactly the numerator.
Put $xe^x=t \Rightarrow e^x(1+x)\,dx=dt$.

$$\int \frac{e^x(1+x)}{\sin^2(xe^x)}dx=\int \operatorname{cosec}^2 t\,dt=-\cot t+c=-\cot(xe^x)+c.$$

Jun 2026 version: $\int \frac{e^x(1+x)}{\cos^2(xe^x)}dx=\int \sec^2 t\,dt=\tan(xe^x)+c.$
Textbook version: $\int \frac{e^x(1+x)}{\cos(xe^x)}dx=\int \sec t\,dt=\log|\sec(xe^x)+\tan(xe^x)|+c.$

Example 4.4 — $\int x^x(1+\log x)\,dx$ [PYQ Feb 2025, MCQ]

Let $t=x^x$. Then $\log t=x\log x$, and differentiating, $\frac{1}{t}\frac{dt}{dx}=1+\log x$, i.e. $dt=x^x(1+\log x)\,dx$. So

$$\int x^x(1+\log x)\,dx=\int dt=t+c=x^x+c.$$

Example 4.5 — $\int \frac{dx}{x+x^{-10}}$ [PYQ Jul 2025, 2 marks]

First clear the negative power: multiply numerator and denominator by x^{10}:

$$\int \frac{x^{10}}{x^{11}+1}dx.$$

Now the numerator is $\frac{1}{11}$ of the derivative of the denominator. Put $x^{11}+1=t$, $11x^{10}dx=dt$:

$$I=\frac{1}{11}\int\frac{dt}{t}=\frac{1}{11}\log|x^{11}+1|+c.$$

Textbook generalisation: $\int \frac{dx}{3x+7x^{-n}}=\int \frac{x^n\,dx}{3x^{n+1}+7}=\frac{1}{3(n+1)}\log|3x^{n+1}+7|+c.$

Example 4.6 — Root inside: $\int \frac{\cos\sqrt{x}}{\sqrt{x}}\,dx$ and $\int \frac{dx}{x+\sqrt{x}}$ (Textbook)

First: $\sqrt{x}=t \Rightarrow \frac{dx}{2\sqrt{x}}=dt \Rightarrow \frac{dx}{\sqrt{x}}=2dt$. So $I=2\int\cos t\,dt=2\sin\sqrt{x}+c$.
Second: $x+\sqrt{x}=\sqrt{x}(\sqrt{x}+1)$. Put $\sqrt{x}+1=t$, $\frac{dx}{2\sqrt{x}}=dt$:

$$\int\frac{dx}{\sqrt{x}(\sqrt{x}+1)}=2\int\frac{dt}{t}=2\log(\sqrt{x}+1)+c.$$

Example 4.7 — Log inside: $\int \frac{\cot(\log x)}{x}\,dx$ and $\int \frac{1+x}{x\sin(x+\log x)}dx$

First (Textbook): $\log x = t$, $\frac{dx}{x} = dt$: $\int \cot t\,dt = \log|\sin t| + c = \log|\sin(\log x)| + c$.
Second (Ex 3.2 A I.3): the inner function is $x + \log x$, with derivative $1 + \frac{1}{x} = \frac{1+x}{x}$ — exactly the rest of the integrand. Put $x + \log x = t$:

$$\int \frac{dt}{\sin t} = \int \operatorname{cosec} t\,dt = \log|\operatorname{cosec} t - \cot t| + c = \log\left|\tan\frac{x+\log x}{2}\right| + c.$$

Example 4.8 — Exponential inside: $\int \frac{1}{1+e^{-x}}\,dx$ and $\int 5^{5^x}5^x\,dx$ (Textbook)

First: multiply by e^x: $\int \frac{e^x}{e^x+1}dx = \log(e^x+1) + c$ (Corollary III).
Second: $5^x = t \Rightarrow 5^x \log 5\,dx = dt$. $I = \frac{1}{\log 5}\int 5^t\,dt = \frac{1}{\log 5}\cdot\frac{5^t}{\log 5} + c = \frac{5^{5^x}}{(\log 5)^2} + c$.

Example 4.9 — Shift substitution: $\int (3x+2)\sqrt{x-4}\,dx$ (Textbook)

Put $x - 4 = t$, so $x = t + 4$ and $dx = dt$. Then $3x + 2 = 3t + 14$:

$$\int (3t+14)t^{1/2}dt = \int \left(3t^{3/2} + 14t^{1/2}\right)dt = \frac{6}{5}t^{5/2} + \frac{28}{3}t^{3/2} + c = \frac{6}{5}(x-4)^{5/2} + \frac{28}{3}(x-4)^{3/2} + c.$$

Example 4.10 — $\int \frac{e^x+1}{e^x-1}\,dx$ (Textbook)

Split: $\frac{e^x+1}{e^x-1} = 1 + \frac{2}{e^x-1} = 1 + \frac{2e^{-x}}{1-e^{-x}}$ (divide top and bottom of the second fraction by e^x). The derivative of $1 - e^{-x}$ is e^{-x}, so

$$I = x + 2\log|1 - e^{-x}| + c.$$

4.4 Integrals of tan, cot, sec, cosec (§3.2.2) — derivations asked as theory

Four standard results

$$\int \tan x\,dx = \log|\sec x| + c, \qquad \int \cot x\,dx = \log|\sin x| + c$$

$$\int \sec x\,dx = \log|\sec x + \tan x| + c = \log\left|\tan\left(\frac{x}{2} + \frac{\pi}{4}\right)\right| + c$$

$$\int \operatorname{cosec} x\,dx = \log|\operatorname{cosec} x - \cot x| + c = \log\left|\tan\frac{x}{2}\right| + c$$

Derivations

tan: $\int \frac{\sin x}{\cos x}dx = -\int \frac{-\sin x}{\cos x}dx = -\log|\cos x| + c = \log|\sec x| + c$ (numerator is $-$(derivative of denominator)).
cot: $\int \frac{\cos x}{\sin x}dx = \log|\sin x| + c$.
sec: multiply and divide by $(\sec x + \tan x)$:

$$\int \sec x\,dx = \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x}dx = \int \frac{\sec x\tan x + \sec^2 x}{\sec x + \tan x}dx.$$

The numerator is exactly $\frac{d}{dx}(\sec x + \tan x)$, so the result is $\log|\sec x + \tan x| + c$.
cosec: multiply and divide by $(\operatorname{cosec} x - \cot x)$:

$$\int \frac{\operatorname{cosec}^2 x - \operatorname{cosec} x\cot x}{\operatorname{cosec} x - \cot x}dx = \int \frac{-\operatorname{cosec} x\cot x + \operatorname{cosec}^2 x}{\operatorname{cosec} x - \cot x}dx = \log|\operatorname{cosec} x - \cot x| + c,$$

since $\frac{d}{dx}(\operatorname{cosec} x - \cot x) = -\operatorname{cosec} x\cot x + \operatorname{cosec}^2 x$.

Common Mistake

Students write $\int \tan x\,dx = \sec^2 x$ (confusing integral with derivative) or $\int \sec x\,dx = \sec x \tan x$. Remember: $\sec x \tan x$ is the **derivative** of $\sec x$, not its integral.

4.5 Special trigonometric types in Ex 3.2 A

Type T1: $\displaystyle\int \frac{\sin(x+a)}{\cos(x-b)}dx$ **— the twice-asked PYQ**

Example 4.11 — $\int \frac{\sin(x+a)}{\cos(x-b)}\,dx$ [PYQ Jul 2024 & Jul 2025, 3 marks]

Key idea: rewrite the numerator's angle using the denominator's angle: $x+a = (x-b)+(a+b)$.

$$\sin(x+a) = \sin[(x-b)+(a+b)] = \sin(x-b)\cos(a+b) + \cos(x-b)\sin(a+b).$$

Divide by $\cos(x-b)$:

$$I = \int [\cos(a+b)\tan(x-b) + \sin(a+b)]\,dx.$$

Here $\cos(a+b)$ and $\sin(a+b)$ are **constants** (no x in them). So

$$I = \cos(a+b)\log|\sec(x-b)| + x\sin(a+b) + c.$$

Marking scheme logic for this question

Writing the split $x+a = (x-b)+(a+b)$ and the expansion earns the first mark; the correct division into tan and constant earns the second; the final integral with $\log|\sec(x-b)|$ earns the third. Do not skip the expansion line.

Type T2: $\displaystyle\int \frac{a\cos x + b\sin x}{c\cos x + d\sin x}dx$ **— "Numerator = λ(Denominator) + μ(Denominator)′"**

Example 4.12 — $\int \frac{dx}{1-\tan x}$ (Textbook) by the λ, μ method

$\int \frac{dx}{1-\tan x} = \int \frac{\cos x}{\cos x - \sin x}dx$. Denominator $D = \cos x - \sin x$, so $D' = -\sin x - \cos x$.
Write $\cos x = \lambda(\cos x - \sin x) + \mu(-\sin x - \cos x)$. Comparing coefficients:

$$\cos x:\ \lambda - \mu = 1, \qquad \sin x:\ -\lambda - \mu = 0 \ \Rightarrow\ \lambda = \tfrac{1}{2},\ \mu = -\tfrac{1}{2}.$$

$$I = \int \left[\lambda + \mu\frac{D'}{D}\right]dx = \frac{1}{2}x - \frac{1}{2}\log|\cos x - \sin x| + c.$$

(The textbook reaches the equivalent form $\frac{1}{2}\left[x + \frac{\pi}{4} + \log\left|\sec\left(x+\frac{\pi}{4}\right)\right|\right] + c$ through a $\cos(x+\pi/4)$ substitution. The two answers differ only by a constant — check by differentiating.)

Example 4.13 — $\int \frac{\sin x + 2\cos x}{3\sin x + 4\cos x}\,dx$ (Ex 3.2 A II.5)

$D = 3\sin x + 4\cos x$, $D' = 3\cos x - 4\sin x$. Let $\sin x + 2\cos x = \lambda(3\sin x + 4\cos x) + \mu(3\cos x - 4\sin x)$.

$$\sin x:\ 3\lambda - 4\mu = 1, \qquad \cos x:\ 4\lambda + 3\mu = 2.$$

Solving: multiply the first by 3 and the second by 4 and add: $25\lambda = 11 \Rightarrow \lambda = \frac{11}{25}$; then $\mu = \frac{3\lambda - 1}{4} = \frac{2}{25}$.

$$I = \frac{11}{25}x + \frac{2}{25}\log|3\sin x + 4\cos x| + c.$$

The same method applies to exponentials: $\int \frac{4e^x - 25}{2e^x - 5}dx$, $\int \frac{20 + 12e^x}{3e^x + 4}dx$ (Ex 3.2 A II.7–9).

Type T3: powers of sin, cos, tan

Strategy for powers

- **Odd power of** sin **or** cos**:** peel off one factor and convert the rest using $\sin^2+\cos^2=1$, then substitute. E.g. $\cos^7 x=(1-\sin^2 x)^3\cos x$, put $\sin x=t$.
- $\sin^m x\cos^n x$ **with one odd power:** substitute the *other* function.
- **Powers of** tan**:** use $\tan^2 x=\sec^2 x-1$ repeatedly; then $\tan x=t$ (since $\sec^2 x\,dx=dt$).
- **Even powers only:** power-reduction formulas (Example 3.4).

Example 4.14 — $\int \sin^4 x\cos^3 x\,dx$ (Ex 3.2 A I.9)

cos has the odd power. $\cos^3 x=(1-\sin^2 x)\cos x$. Put $\sin x=t$, $\cos x\,dx=dt$:

$$\int t^4(1-t^2)\,dt=\frac{t^5}{5}-\frac{t^7}{7}+c=\frac{\sin^5 x}{5}-\frac{\sin^7 x}{7}+c.$$

Example 4.15 — $\int \tan^5 x\,dx$ (Ex 3.2 A II.11)

$\tan^5 x=\tan^3 x(\sec^2 x-1)=\tan^3 x\sec^2 x-\tan x(\sec^2 x-1)=\tan^3 x\sec^2 x-\tan x\sec^2 x+\tan x$.

$$\int \tan^5 x\,dx=\frac{\tan^4 x}{4}-\frac{\tan^2 x}{2}+\log|\sec x|+c.$$

(First two terms by $\tan x=t$; last by the standard result.)

Example 4.16 — $\int \frac{\sec^8 x}{\operatorname{cosec} x}\,dx$ (Textbook)

$\frac{\sec^8 x}{\operatorname{cosec} x}=\sec^7 x\cdot\sec x\cdot\sin x=\sec^7 x\cdot\tan x=\sec^6 x\,(\sec x\tan x)$. Put $\sec x=t$, $\sec x\tan x\,dx=dt$:

$$\int t^6\,dt=\frac{t^7}{7}+c=\frac{\sec^7 x}{7}+c.$$

Example 4.17 — $\int \tan 3x\tan 2x\tan x\,dx$ (Ex 3.2 A II.13) — the identity trick

Since $3x=2x+x$: $\tan 3x=\frac{\tan 2x+\tan x}{1-\tan 2x\tan x}$, so $\tan 3x-\tan 3x\tan 2x\tan x=\tan 2x+\tan x$, giving

$$\tan 3x\tan 2x\tan x=\tan 3x-\tan 2x-\tan x.$$

$$I=\frac{1}{3}\log|\sec 3x|-\frac{1}{2}\log|\sec 2x|-\log|\sec x|+c.$$

5. Seven Special Integrals (§3.2.3) — Proofs Are Board Questions

The seven formulas

1. $\displaystyle\int\frac{dx}{x^2+a^2}=\frac{1}{a}\tan^{-1}\frac{x}{a}+c$
2. $\displaystyle\int\frac{dx}{x^2-a^2}=\frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|+c$
3. $\displaystyle\int\frac{dx}{a^2-x^2}=\frac{1}{2a}\log\left|\frac{a+x}{a-x}\right|+c$
4. $\displaystyle\int\frac{dx}{\sqrt{a^2-x^2}}=\sin^{-1}\frac{x}{a}+c$
5. $\displaystyle\int\frac{dx}{\sqrt{x^2-a^2}}=\log\left|x+\sqrt{x^2-a^2}\right|+c$
6. $\displaystyle\int\frac{dx}{\sqrt{x^2+a^2}}=\log\left|x+\sqrt{x^2+a^2}\right|+c$
7. $\displaystyle\int\frac{dx}{x\sqrt{x^2-a^2}}=\frac{1}{a}\sec^{-1}\frac{x}{a}+c$

How to tell them apart

- **No root**: $x^2 + a^2 \to \tan^{-1}$ (sum never factorises); $x^2 - a^2$ or $a^2 - x^2 \to \log$ (difference factorises, partial fractions give logs).
- **With root**: $\sqrt{a^2 - x^2} \to \sin^{-1}$ (the only one where x^2 is *subtracted*); $\sqrt{x^2 \pm a^2} \to \log|x + \sqrt{\cdot}|$ (the root is copied inside the log).
- For formulas 2 and 3, the **positive** term goes first in the numerator of the log: $x^2 - a^2 \to \frac{x-a}{x+a}$; $a^2 - x^2 \to \frac{a+x}{a-x}$.

Proof of Formula 1 (substitution $x = a\tan\theta$)

Put $x = a\tan\theta$, so $dx = a\sec^2\theta\, d\theta$ and $\theta = \tan^{-1}\frac{x}{a}$.

$$I = \int \frac{a\sec^2\theta\, d\theta}{a^2\tan^2\theta + a^2} = \int \frac{a\sec^2\theta}{a^2\sec^2\theta} d\theta = \frac{1}{a}\int d\theta = \frac{1}{a}\theta + c = \frac{1}{a}\tan^{-1}\frac{x}{a} + c.$$

Proof of Formula 2 (partial fractions)

$x^2 - a^2 = (x-a)(x+a)$ and $\dfrac{1}{(x-a)(x+a)} = \dfrac{1}{2a}\left[\dfrac{1}{x-a} - \dfrac{1}{x+a}\right]$ (check by combining the right side). So

$$I = \frac{1}{2a}\left[\log|x-a| - \log|x+a|\right] + c = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + c.$$

Proof of Formula 3 [asked as PYQ Mar 2024, 3 marks]

$a^2 - x^2 = (a-x)(a+x)$. Resolve into partial fractions:

$$\frac{1}{(a-x)(a+x)} = \frac{A}{a-x} + \frac{B}{a+x} \;\Rightarrow\; 1 = A(a+x) + B(a-x).$$

Put $x = a$: $1 = 2aA \Rightarrow A = \frac{1}{2a}$. Put $x = -a$: $1 = 2aB \Rightarrow B = \frac{1}{2a}$. Hence

$$I = \frac{1}{2a}\int\left[\frac{1}{a-x} + \frac{1}{a+x}\right]dx = \frac{1}{2a}\left[\frac{\log|a-x|}{-1} + \log|a+x|\right] + c = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + c.$$

Watch the (-1): $\int \frac{dx}{a-x} = -\log|a-x|$, because the coefficient of x is -1.

Proof of Formula 4 (substitution $x = a\sin\theta$)

$dx = a\cos\theta\, d\theta$, $\theta = \sin^{-1}\frac{x}{a}$, and $\sqrt{a^2 - a^2\sin^2\theta} = a\cos\theta$.

$$I = \int \frac{a\cos\theta}{a\cos\theta} d\theta = \theta + c = \sin^{-1}\frac{x}{a} + c.$$

Proof of Formula 5 (substitution $x = a\sec\theta$)

$dx = a\sec\theta\tan\theta\, d\theta$, and $\sqrt{a^2\sec^2\theta - a^2} = a\tan\theta$.

$$I = \int \frac{a\sec\theta\tan\theta}{a\tan\theta} d\theta = \int \sec\theta\, d\theta = \log|\sec\theta + \tan\theta| + c_1.$$

Now $\sec\theta = \frac{x}{a}$ and $\tan\theta = \sqrt{\sec^2\theta - 1} = \frac{\sqrt{x^2-a^2}}{a}$, so

$$I = \log\left|\frac{x + \sqrt{x^2-a^2}}{a}\right| + c_1 = \log\left|x + \sqrt{x^2-a^2}\right| - \log a + c_1 = \log\left|x + \sqrt{x^2-a^2}\right| + c,$$

where $c = c_1 - \log a$ (a constant absorbed into a constant).

Proof of Formula 6 (substitution $x = a\tan\theta$)

$dx = a\sec^2\theta\,d\theta$ and $\sqrt{a^2\tan^2\theta + a^2} = a\sec\theta$. So $I = \int \sec\theta\,d\theta = \log|\sec\theta + \tan\theta| + c_1$. With $\tan\theta = \frac{x}{a}$, $\sec\theta = \frac{\sqrt{x^2+a^2}}{a}$, exactly as in Formula 5 this gives $\log\left|x + \sqrt{x^2+a^2}\right| + c$.

Proof of Formula 7 (substitution $x = a\sec\theta$)

$dx = a\sec\theta\tan\theta\,d\theta$; $x\sqrt{x^2-a^2} = a\sec\theta \cdot a\tan\theta$.

$$I = \int \frac{a\sec\theta\tan\theta}{a^2\sec\theta\tan\theta}d\theta = \frac{1}{a}\int d\theta = \frac{1}{a}\theta + c = \frac{1}{a}\sec^{-1}\frac{x}{a} + c.$$

Example 5.1 — Direct use: $\int \frac{dx}{x^2+25}$ [PYQ Mar 2024, 1 mark], $\int \frac{dx}{25-9x^2}$ [PYQ Feb 2025, 2 marks]

First: $a = 5$: $\int \frac{dx}{x^2+5^2} = \frac{1}{5}\tan^{-1}\frac{x}{5} + c$.
Second: make the coefficient of x^2 equal to 1 by taking 9 out:

$$\int \frac{dx}{9\left(\frac{25}{9} - x^2\right)} = \frac{1}{9}\int \frac{dx}{\left(\frac{5}{3}\right)^2 - x^2} = \frac{1}{9}\cdot\frac{1}{2\cdot\frac{5}{3}}\log\left|\frac{\frac{5}{3}+x}{\frac{5}{3}-x}\right| + c = \frac{1}{30}\log\left|\frac{5+3x}{5-3x}\right| + c.$$

Common Mistake — the leading coefficient

In $\int \frac{dx}{25-9x^2}$, writing $a = 5$ and "x" $= 3x$ but forgetting the $\frac{1}{3}$ from $d(3x) = 3\,dx$. Either take the 9 out (as above) or put $3x = t$ and keep $dx = \frac{dt}{3}$. Both routes must give $\frac{1}{30}$.

6. Quadratic Expressions — Completing the Square (§3.2.4)

Why complete the square?

The seven formulas only have $x^2 \pm a^2$ or $a^2 - x^2$. A general quadratic $ax^2 + bx + c$ has an extra bx term. Completing the square turns it into $a[(x+\beta)^2 \pm \alpha^2]$, i.e. *a shifted* $x^2 \pm \alpha^2$. Since shifting x by a constant does not change dx, the same formula applies with x replaced by $(x+\beta)$.

Procedure for $\int \frac{dx}{ax^2+bx+c}$ and $\int \frac{dx}{\sqrt{ax^2+bx+c}}$

1. Take a (or $\sqrt{a}$ for the root form) outside: $a\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right)$. If $a < 0$, take $-a$ outside and write $\left(\frac{-c}{a}\ldots\right) - \left(x^2\ldots\right)$.
2. Add and subtract $\left(\frac{1}{2} \times \text{coefficient of } x\right)^2$.
3. Write as $(x+\beta)^2 \pm \alpha^2$ or $\alpha^2 - (x+\beta)^2$.
4. Apply the matching special formula with $x \to x + \beta$.

Example 6.1 — $\int \frac{dx}{x^2+8x+12}$ (Textbook)

$\left(\frac{1}{2}\cdot 8\right)^2 = 16$. $x^2 + 8x + 12 = (x^2 + 8x + 16) - 4 = (x+4)^2 - 2^2$. Formula 2 with $x \to x+4$, $a = 2$:

$$I = \frac{1}{2\cdot 2}\log\left|\frac{x+4-2}{x+4+2}\right| + c = \frac{1}{4}\log\left|\frac{x+2}{x+6}\right| + c.$$

Example 6.2 — $\int \frac{dx}{\sqrt{3x^2-4x+2}}$ (Textbook)

$\sqrt{3x^2-4x+2} = \sqrt{3}\sqrt{x^2 - \frac{4}{3}x + \frac{2}{3}}$. Half of $-\frac{4}{3}$ is $-\frac{2}{3}$; square $= \frac{4}{9}$.

$$x^2 - \frac{4}{3}x + \frac{2}{3} = \left(x - \frac{2}{3}\right)^2 + \frac{2}{3} - \frac{4}{9} = \left(x - \frac{2}{3}\right)^2 + \left(\frac{\sqrt{2}}{3}\right)^2.$$

Formula 6:

$$I = \frac{1}{\sqrt{3}} \log \left| x - \frac{2}{3} + \sqrt{x^2 - \frac{4}{3}x + \frac{2}{3}} \right| + c.$$

Example 6.3 — Negative leading coefficient: $\int \frac{dx}{3-10x-25x^2}$ (Textbook) — with a textbook correction

$3 - 10x - 25x^2 = 3 - (25x^2 + 10x) = 3 - [(5x+1)^2 - 1] = 4 - (5x+1)^2 = 2^2 - (5x+1)^2$.
Put $5x + 1 = t$, $dx = \frac{dt}{5}$, and use Formula 3 with $a = 2$:

$$I = \frac{1}{5} \int \frac{dt}{2^2 - t^2} = \frac{1}{5} \cdot \frac{1}{4} \log \left| \frac{2+t}{2-t} \right| + c = \frac{1}{20} \log \left| \frac{3+5x}{1-5x} \right| + c.$$

Textbook erratum (verified)

The textbook (p. 117, solved Ex. 6) prints the answer $\frac{1}{5} \log \left| \frac{1+5x}{3-5x} \right| + c$. In its working it completes the square as $\left(x - \frac{1}{5}\right)^2$ instead of $\left(x + \frac{1}{5}\right)^2$ (the coefficient of x inside the bracket is $+\frac{2}{5}$), and the final constant $\frac{1}{5}$ should be $\frac{1}{20}$. You can confirm: differentiating $\frac{1}{20} \log \left| \frac{3+5x}{1-5x} \right|$ gives exactly $\frac{1}{3-10x-25x^2}$, while differentiating the printed answer does not. In an exam, write the method; your correct answer will be marked correct.

Example 6.4 — $\int \frac{dx}{\sqrt{1+x-x^2}}$ (Textbook)

$1 + x - x^2 = 1 - \left(x^2 - x + \frac{1}{4}\right) + \frac{1}{4} = \frac{5}{4} - \left(x - \frac{1}{2}\right)^2 = \left(\frac{\sqrt{5}}{2}\right)^2 - \left(x - \frac{1}{2}\right)^2$. Formula 4:

$$I = \sin^{-1} \frac{x - \frac{1}{2}}{\frac{\sqrt{5}}{2}} + c = \sin^{-1} \frac{2x-1}{\sqrt{5}} + c.$$

Example 6.5 — Substitution first, then a quadratic: $\int \frac{\sin 2x}{3\sin^4 x - 4\sin^2 x + 1}\,dx$ (Textbook)

Put $\sin^2 x = t$, $2 \sin x \cos x\, dx = \sin 2x\, dx = dt$. The integral becomes $\int \frac{dt}{3t^2 - 4t + 1}$, and $3t^2 - 4t + 1 = (3t-1)(t-1)$. Partial fractions (or complete the square): $\frac{1}{(3t-1)(t-1)} = \frac{-3/2}{3t-1} + \frac{1/2}{t-1}$. Then

$$I = -\frac{1}{2} \log|3t-1| + \frac{1}{2} \log|t-1| + c = \frac{1}{2} \log \left| \frac{\sin^2 x - 1}{3\sin^2 x - 1} \right| + c.$$

(The textbook writes $\frac{1}{2} \log \left| \frac{3\sin^2 x - 3}{3\sin^2 x - 1} \right|$; the extra factor 3 inside the log is just a constant $\frac{1}{2}\log 3$ absorbed into c.)

Example 6.6 — $\int \frac{e^{x/2}}{\sqrt{e^{-x} - e^x}}\,dx$ (Textbook)

$\sqrt{e^{-x} - e^x} = \sqrt{\frac{1 - e^{2x}}{e^x}} = \frac{\sqrt{1 - (e^x)^2}}{e^{x/2}}$. So the integrand is $\frac{e^{x/2} \cdot e^{x/2}}{\sqrt{1-(e^x)^2}} = \frac{e^x}{\sqrt{1-(e^x)^2}}$. Put $e^x = t$:

$$\int \frac{dt}{\sqrt{1 - t^2}} = \sin^{-1} t + c = \sin^{-1}(e^x) + c.$$

Example 6.7 — The hardest textbook substitution: $\int (\sqrt{\tan x} + \sqrt{\cot x})\,dx$

$\sqrt{\tan x} + \sqrt{\cot x} = \frac{\tan x + 1}{\sqrt{\tan x}}$. Put $\sqrt{\tan x} = t$: $\tan x = t^2$, $\sec^2 x\, dx = 2t\, dt$, so $dx = \frac{2t\,dt}{1+t^4}$.

$$I = \int \frac{t^2+1}{t} \cdot \frac{2t}{1+t^4} dt = 2\int \frac{t^2+1}{t^4+1} dt = 2\int \frac{1 + \frac{1}{t^2}}{t^2 + \frac{1}{t^2}} dt = 2\int \frac{1 + \frac{1}{t^2}}{\left(t - \frac{1}{t}\right)^2 + 2} dt.$$

Put $t - \frac{1}{t} = u$, $\left(1 + \frac{1}{t^2}\right) dt = du$:

$$I = 2\int \frac{du}{u^2 + (\sqrt{2})^2} = \sqrt{2}\tan^{-1}\frac{u}{\sqrt{2}} + c = \sqrt{2}\tan^{-1}\left(\frac{\tan x - 1}{\sqrt{2}\tan x}\right) + c.$$

7. Trigonometric Denominators (§3.2.5 and §3.2.6)

7.1 Type $\int \frac{dx}{a\sin^2 x + b\cos^2 x + c}$ — divide by $\cos^2 x$

Procedure

1. Divide numerator and denominator by $\cos^2 x$. Numerator becomes $\sec^2 x$.
2. In the denominator, replace $\sec^2 x$ (coming from the constant c) by $1 + \tan^2 x$.
3. Put $\tan x = t$, $\sec^2 x\, dx = dt$. You get $\int \frac{dt}{At^2 + B}$ — a special integral.

Example 7.1 — $\int \frac{2\,dx}{3+2\sin^2 x + 5\cos^2 x}$ [PYQ Jun 2026, 4 marks; textbook solved Ex. 15]

Divide by $\cos^2 x$:

$$I = \int \frac{2\sec^2 x}{3\sec^2 x + 2\tan^2 x + 5}dx = \int \frac{2\sec^2 x}{3(1+\tan^2 x) + 2\tan^2 x + 5}dx = \int \frac{2\sec^2 x}{5\tan^2 x + 8}dx.$$

Put $\tan x = t$, $\sec^2 x\, dx = dt$:

$$I = 2\int \frac{dt}{5t^2 + 8} = \frac{2}{5}\int \frac{dt}{t^2 + \left(\sqrt{\frac{8}{5}}\right)^2} = \frac{2}{5}\cdot\frac{1}{\sqrt{8/5}}\tan^{-1}\frac{t}{\sqrt{8/5}} + c.$$

Simplify: $\sqrt{\frac{8}{5}} = \frac{2\sqrt{2}}{\sqrt{5}}$, so $\frac{2}{5}\cdot\frac{\sqrt{5}}{2\sqrt{2}} = \frac{1}{\sqrt{10}}$:

$$I = \frac{1}{\sqrt{10}}\tan^{-1}\left(\frac{\sqrt{5}\tan x}{2\sqrt{2}}\right) + c.$$

(Without the factor 2 in the numerator — the textbook's version — the answer is $\frac{1}{2\sqrt{10}}\tan^{-1}\left(\frac{\sqrt{5}\tan x}{2\sqrt{2}}\right) + c$.)

Example 7.2 — $\int \frac{\sin x}{\sin 3x}\,dx$ [PYQ Feb 2026, 3 marks]

Use $\sin 3x = 3\sin x - 4\sin^3 x = \sin x(3 - 4\sin^2 x)$:

$$I = \int \frac{dx}{3 - 4\sin^2 x}.$$

Divide by $\cos^2 x$: $I = \int \frac{\sec^2 x}{3\sec^2 x - 4\tan^2 x}dx = \int \frac{\sec^2 x}{3 + 3\tan^2 x - 4\tan^2 x}dx = \int \frac{\sec^2 x}{3 - \tan^2 x}dx.$

Put $\tan x = t$:

$$I = \int \frac{dt}{(\sqrt{3})^2 - t^2} = \frac{1}{2\sqrt{3}}\log\left|\frac{\sqrt{3} + \tan x}{\sqrt{3} - \tan x}\right| + c.$$

Twin (Textbook Ex. 16): $\int \frac{\cos\theta}{\cos 3\theta}d\theta = \int \frac{d\theta}{4\cos^2\theta - 3}$; dividing by $\cos^2\theta$ gives $\int \frac{\sec^2\theta}{1 - 3\tan^2\theta}d\theta = \frac{1}{2\sqrt{3}}\log\left|\frac{1+\sqrt{3}\tan\theta}{1-\sqrt{3}\tan\theta}\right| + c$.

7.2 Type $\displaystyle\int \frac{dx}{a\sin x + b\cos x + c}$ — the $t = \tan\frac{x}{2}$ substitution

Substitution kit

$$t = \tan\frac{x}{2}, \quad dx = \frac{2\,dt}{1+t^2}, \quad \sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}.$$

For $\sin 2x,\ \cos 2x$ use $t = \tan x$: $dx = \frac{dt}{1+t^2}$, $\sin 2x = \frac{2t}{1+t^2}$, $\cos 2x = \frac{1-t^2}{1+t^2}$.

Example 7.3 — $\int \frac{dx}{5-4\cos x}$ (Textbook)

$$I = \int \frac{\frac{2\,dt}{1+t^2}}{5 - 4\cdot\frac{1-t^2}{1+t^2}} = \int \frac{2\,dt}{5(1+t^2) - 4(1-t^2)} = \int \frac{2\,dt}{9t^2+1} = \frac{2}{9}\int \frac{dt}{t^2 + \left(\frac{1}{3}\right)^2}.$$

$$I = \frac{2}{9}\cdot 3\tan^{-1}(3t) + c = \frac{2}{3}\tan^{-1}\left(3\tan\frac{x}{2}\right) + c.$$

Example 7.4 — $\int \frac{dx}{3-2\sin x+5\cos x}$ (Textbook)

Denominator after multiplying by $(1+t^2)$: $3(1+t^2) - 2(2t) + 5(1-t^2) = 8 - 4t - 2t^2$.

$$I = \int \frac{2\,dt}{8-4t-2t^2} = \int \frac{dt}{4-2t-t^2} = \int \frac{dt}{5-(t+1)^2} = \frac{1}{2\sqrt{5}}\log\left|\frac{\sqrt{5}+1+\tan\frac{x}{2}}{\sqrt{5}-1-\tan\frac{x}{2}}\right| + c.$$

Example 7.5 — $\int \frac{dx}{2-3\sin 2x}$ (Textbook) — use $t = \tan x$

$$I = \int \frac{\frac{dt}{1+t^2}}{2 - \frac{6t}{1+t^2}} = \int \frac{dt}{2t^2-6t+2} = \frac{1}{2}\int \frac{dt}{\left(t-\frac{3}{2}\right)^2 - \left(\frac{\sqrt{5}}{2}\right)^2} = \frac{1}{2\sqrt{5}}\log\left|\frac{2\tan x - 3 - \sqrt{5}}{2\tan x - 3 + \sqrt{5}}\right| + c.$$

Example 7.6 — $\int \frac{dx}{\sin x - \sqrt{3}\cos x}$ (Textbook, shortcut method)

Write $\sin x - \sqrt{3}\cos x = 2\left(\frac{1}{2}\sin x - \frac{\sqrt{3}}{2}\cos x\right) = 2\sin\left(x - \frac{\pi}{3}\right)$. Then

$$I = \frac{1}{2}\int \operatorname{cosec}\left(x - \frac{\pi}{3}\right) dx = \frac{1}{2}\log\left|\tan\left(\frac{x}{2} - \frac{\pi}{6}\right)\right| + c.$$

Rule: when there is no constant term ($c = 0$), converting $a\sin x + b\cos x$ into $R\sin(x \pm \alpha)$ is faster than $t = \tan\frac{x}{2}$.

8. Linear over Quadratic (§3.2.6, Ex 3.2 C)

Method: Numerator = $A \cdot \frac{d}{dx}$(Denominator quadratic) $+B$

For $\displaystyle\int \frac{px+q}{ax^2+bx+c}dx$ or $\displaystyle\int \frac{px+q}{\sqrt{ax^2+bx+c}}dx$:

1. Write $px + q = A\,(2ax + b) + B$. Compare coefficients of x and constants to find A, B.
2. Split: $A\int \frac{D'}{D}dx + B\int \frac{dx}{D}$ (or $A\int \frac{D'}{\sqrt{D}}dx + B\int \frac{dx}{\sqrt{D}}$).
3. First part: $A\log|D|$ (or $2A\sqrt{D}$). Second part: complete the square.

Example 8.1 — $\int \frac{2x-3}{3x^2+4x+5}\,dx$ (Textbook)

$D' = 6x+4$. $2x-3 = A(6x+4)+B \Rightarrow 6A = 2,\ 4A+B = -3 \Rightarrow A = \frac{1}{3},\ B = -\frac{13}{3}$.
$I_1 = \frac{1}{3}\int \frac{6x+4}{3x^2+4x+5}dx = \frac{1}{3}\log(3x^2+4x+5)$.
$I_2 = \frac{13}{3}\cdot\frac{1}{3}\int \frac{dx}{x^2+\frac{4}{3}x+\frac{5}{3}} = \frac{13}{9}\int \frac{dx}{\left(x+\frac{2}{3}\right)^2+\left(\frac{\sqrt{11}}{3}\right)^2} = \frac{13}{9}\cdot\frac{3}{\sqrt{11}}\tan^{-1}\frac{3x+2}{\sqrt{11}} = \frac{13}{3\sqrt{11}}\tan^{-1}\frac{3x+2}{\sqrt{11}}$.

$$I = \frac{1}{3}\log(3x^2+4x+5) - \frac{13}{3\sqrt{11}}\tan^{-1}\left(\frac{3x+2}{\sqrt{11}}\right)+c.$$

Example 8.2 — $\int \sqrt{\frac{x-5}{x-7}}\,dx$ (Textbook)

Multiply numerator and denominator inside the root by $(x-5)$:

$$\sqrt{\frac{(x-5)^2}{(x-7)(x-5)}} = \frac{x-5}{\sqrt{x^2-12x+35}}.$$

$D' = 2x-12$; $x-5 = \frac{1}{2}(2x-12)+1$, so $A = \frac{1}{2}, B = 1$.

$$\begin{aligned} I &= \frac{1}{2}\int \frac{2x-12}{\sqrt{x^2-12x+35}}dx + \int \frac{dx}{\sqrt{(x-6)^2-1}} \\ &= \sqrt{x^2-12x+35} + \log\left|x-6+\sqrt{x^2-12x+35}\right| + c. \end{aligned}$$

(Similarly $\int \sqrt{\frac{8-x}{x}}dx = \sqrt{8x-x^2} + 4\sin^{-1}\frac{x-4}{4} + c$.)

Example 8.3 — $\int \frac{3\cos x}{4\sin^2 x+4\sin x-1}\,dx$ (Ex 3.2 C I.8)

Put $\sin x = t$: $3\int \frac{dt}{4t^2+4t-1} = \frac{3}{4}\int \frac{dt}{\left(t+\frac{1}{2}\right)^2-\left(\frac{1}{\sqrt{2}}\right)^2}$.

$$I = \frac{3}{4}\cdot\frac{1}{2\cdot\frac{1}{\sqrt{2}}}\log\left|\frac{t+\frac{1}{2}-\frac{1}{\sqrt{2}}}{t+\frac{1}{2}+\frac{1}{\sqrt{2}}}\right| + c = \frac{3}{4\sqrt{2}}\log\left|\frac{2\sin x+1-\sqrt{2}}{2\sin x+1+\sqrt{2}}\right| + c.$$

9. Integration by Parts (§3.3, Ex 3.3)

Intuition — the product rule run backwards

The product rule says $\frac{d}{dx}(uw) = u\,w' + w\,u'$. Integrating both sides and rearranging gives a way to trade a hard integral $\int u\,w'\,dx$ for an easier one $\int w\,u'\,dx$. You *differentiate* one factor (it should get simpler, e.g. $x^2 \to 2x \to 2 \to 0$) and *integrate* the other (it should not get harder).

Theorem (§3.3.1) — Integration by parts

If u and v are differentiable functions of x, then

$$\int u\,v\,dx = u\int v\,dx - \int\left[\frac{du}{dx}\int v\,dx\right]dx.$$

In words: **first × integral of second − ∫(derivative of first × integral of second).**

Proof

Let $\int v\,dx = w$, so $v = \frac{dw}{dx}$. By the product rule,

$$\frac{d}{dx}(uw) = u\frac{dw}{dx} + w\frac{du}{dx} = uv + w\frac{du}{dx}.$$

By the definition of integration, $uw = \int \left(uv + w\frac{du}{dx}\right) dx = \int uv\,dx + \int \frac{du}{dx}\,w\,dx$. Substituting $w = \int v\,dx$ and rearranging:

$$\int uv\,dx = u\int v\,dx - \int \left[\frac{du}{dx}\int v\,dx\right] dx.$$

LIATE — which function is the 'first' (u)?

Choose u as the function that comes **first** in: **L**ogarithmic → **I**nverse trig → **A**lgebraic → **T**rigonometric → **E**xponential.

Why it works: functions early in the list become *simpler* when differentiated ($\log x \to \frac{1}{x}$, $\tan^{-1} x \to \frac{1}{1+x^2}$, $x^n \to nx^{n-1}$), while sin, cos and e^x stay just as easy to integrate. The textbook shows what happens with the wrong choice in $\int xe^x\,dx$: taking $u = e^x$ produces $\int x^2e^x\,dx$, which is *harder*.

Example 9.1 — $\int \log x\,dx$ [PYQ Mar 2024, 2 marks]

There is only one function. Write it as $\log x \cdot 1$ and take $u = \log x$ (L), $v = 1$:

$$I = \log x\int 1\,dx - \int \left[\frac{d}{dx}(\log x)\int 1\,dx\right] dx = x\log x - \int \frac{1}{x}\cdot x\,dx = x\log x - x + c.$$

Same trick for $\int \sin^{-1} x\,dx$, $\int \tan^{-1} x\,dx$, $\int (\log x)^2 dx$.

Example 9.2 — $\int x\log x\,dx$ [PYQ Jun 2026, 2 marks]

LIATE: $\log x$ (L) before x (A), so $u = \log x$, $v = x$.

$$I = \log x\cdot\frac{x^2}{2} - \int \frac{1}{x}\cdot\frac{x^2}{2}dx = \frac{x^2}{2}\log x - \frac{1}{2}\int x\,dx = \frac{x^2}{2}\log x - \frac{x^2}{4} + c.$$

Example 9.3 — Repeated parts: $\int x^2\sin 3x\,dx$ [PYQ Jul 2024, 4 marks]

$u = x^2$ (A), $v = \sin 3x$ (T). $\int \sin 3x\,dx = -\frac{\cos 3x}{3}$.

$$I = x^2\left(-\frac{\cos 3x}{3}\right) - \int 2x\left(-\frac{\cos 3x}{3}\right) dx = -\frac{x^2\cos 3x}{3} + \frac{2}{3}\int x\cos 3x\,dx.$$

Parts again on $\int x\cos 3x\,dx$ with $u = x$:

$$\int x\cos 3x\,dx = x\frac{\sin 3x}{3} - \int 1\cdot\frac{\sin 3x}{3}dx = \frac{x\sin 3x}{3} + \frac{\cos 3x}{9}.$$

$$\therefore\ I = -\frac{x^2\cos 3x}{3} + \frac{2}{3}\left(\frac{x\sin 3x}{3} + \frac{\cos 3x}{9}\right) + c = -\frac{x^2\cos 3x}{3} + \frac{2x\sin 3x}{9} + \frac{2\cos 3x}{27} + c.$$

Example 9.4 — Inverse trig × algebraic: $\int x\tan^{-1} x\,dx$ (Textbook)

$u = \tan^{-1} x$ (I), $v = x$.

$$I = \tan^{-1} x\cdot\frac{x^2}{2} - \int \frac{1}{1+x^2}\cdot\frac{x^2}{2}dx = \frac{x^2}{2}\tan^{-1} x - \frac{1}{2}\int \left(1 - \frac{1}{1+x^2}\right) dx$$

$$= \frac{x^2}{2}\tan^{-1}x - \frac{x}{2} + \frac{1}{2}\tan^{-1}x + c.$$

Note the step $\frac{x^2}{1+x^2} = \frac{(1+x^2)-1}{1+x^2} = 1 - \frac{1}{1+x^2}$ — add and subtract to divide.

Example 9.5 — $\int \frac{x}{1-\sin x}\,dx$ (Textbook)

First simplify: $\frac{1}{1-\sin x} = \sec^2 x + \sec x\tan x$ (Example 3.7). Then by parts with $u = x$:

$$I = x(\tan x + \sec x) - \int (\tan x + \sec x)\,dx = x(\tan x + \sec x) - \log|\sec x| - \log|\sec x + \tan x| + c.$$

Since $\sec x(\sec x + \tan x) = \frac{1+\sin x}{\cos^2 x} = \frac{1}{1-\sin x}$, this equals $x(\sec x + \tan x) + \log|1 - \sin x| + c$.

9.1 Cyclic integrals: $\int e^{ax}\sin bx\,dx$, $\int e^{ax}\cos bx\,dx$

Example 9.6 — $\int e^{2x}\sin 3x\,dx$ (Textbook)

Here either function may be the first; apply parts **twice keeping the same choice** ($u = e^{2x}$ both times). The original integral reappears.

$$I = e^{2x}\left(-\frac{\cos 3x}{3}\right) + \frac{2}{3}\int e^{2x}\cos 3x\,dx = -\frac{1}{3}e^{2x}\cos 3x + \frac{2}{3}\left[e^{2x}\frac{\sin 3x}{3} - \frac{2}{3}\int e^{2x}\sin 3x\,dx\right]$$

$$I = -\frac{1}{3}e^{2x}\cos 3x + \frac{2}{9}e^{2x}\sin 3x - \frac{4}{9}I.$$

Bring $\frac{4}{9}I$ across: $\frac{13}{9}I = \frac{e^{2x}}{9}(2\sin 3x - 3\cos 3x)$, so

$$I = \frac{e^{2x}}{13}(2\sin 3x - 3\cos 3x) + c.$$

General results (textbook activity — memorise)

$$\int e^{ax}\sin(bx+c)\,dx = \frac{e^{ax}}{a^2+b^2}\left[a\sin(bx+c) - b\cos(bx+c)\right] + k$$

$$\int e^{ax}\cos(bx+c)\,dx = \frac{e^{ax}}{a^2+b^2}\left[a\cos(bx+c) + b\sin(bx+c)\right] + k$$

(The textbook prints the cosine result with "$a\sin + b\cos$", which is a misprint; differentiate to confirm the version above. E.g. $\int e^{2x}\cos 3x\,dx = \frac{e^{2x}}{13}(2\cos 3x + 3\sin 3x) + c$.)

Common Mistake in cyclic integrals

Switching the choice of u in the second application (taking $u = \cos 3x$ the second time). That just undoes the first step and gives $I = I$. Keep the same type of function as u both times.

Example 9.7 — $\int \sec^3 x\,dx$ (Ex 3.3 I.7)

Write $\sec^3 x = \sec x \cdot \sec^2 x$, take $u = \sec x$, $v = \sec^2 x$:

$$I = \sec x\tan x - \int \sec x\tan x\cdot\tan x\,dx = \sec x\tan x - \int \sec x(\sec^2 x - 1)\,dx$$

$$= \sec x\tan x - I + \int \sec x\,dx.$$

$$2I = \sec x\tan x + \log|\sec x + \tan x| \;\Rightarrow\; I = \frac{1}{2}\sec x\tan x + \frac{1}{2}\log|\sec x + \tan x| + c.$$

9.2 The three root formulas (§3.3.1, solved Ex. 6–8)

Results

$$\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + c$$

$$\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right| + c$$

$$\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right| + c$$

Proof of $\int \sqrt{a^2-x^2}\,dx$ [PYQ Jul 2025, 4 marks]

Let $I = \int \sqrt{a^2-x^2}\cdot 1\,dx$. By parts with $u = \sqrt{a^2-x^2}$, $v = 1$:

$$I = \sqrt{a^2-x^2}\cdot x - \int \frac{-2x}{2\sqrt{a^2-x^2}}\cdot x\,dx = x\sqrt{a^2-x^2} + \int \frac{x^2}{\sqrt{a^2-x^2}}dx.$$

Write $x^2 = a^2 - (a^2-x^2)$:

$$\int \frac{x^2}{\sqrt{a^2-x^2}}dx = a^2\int \frac{dx}{\sqrt{a^2-x^2}} - \int \frac{a^2-x^2}{\sqrt{a^2-x^2}}dx = a^2\sin^{-1}\frac{x}{a} - \int \sqrt{a^2-x^2}\,dx = a^2\sin^{-1}\frac{x}{a} - I.$$

Hence $I = x\sqrt{a^2-x^2} + a^2\sin^{-1}\frac{x}{a} - I$, so $2I = x\sqrt{a^2-x^2} + a^2\sin^{-1}\frac{x}{a} + c_1$ and

$$I = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + c \qquad \left(c = \tfrac{c_1}{2}\right).$$

The other two proofs are identical in structure: $x^2 = (x^2 \pm a^2) \mp a^2$, and Formula 6 or 5 replaces $\sin^{-1}$.

Example 9.8 — $\int \sqrt{4+3x-2x^2}\,dx$ (Textbook)

$4+3x-2x^2 = 2\left[2+\frac{3}{2}x-x^2\right] = 2\left[\frac{41}{16} - \left(x-\frac{3}{4}\right)^2\right]$, i.e. $a = \frac{\sqrt{41}}{4}$, inner variable $x-\frac{3}{4}$.

$$I = \sqrt{2}\left[\frac{x-\frac{3}{4}}{2}\sqrt{\frac{41}{16}-\left(x-\frac{3}{4}\right)^2} + \frac{41}{32}\sin^{-1}\frac{x-\frac{3}{4}}{\frac{\sqrt{41}}{4}}\right] + c$$

$$= \frac{4x-3}{8}\sqrt{4+3x-2x^2} + \frac{41}{16\sqrt{2}}\sin^{-1}\left(\frac{4x-3}{\sqrt{41}}\right) + c.$$

(Using $\sqrt{2}\sqrt{\frac{41}{16}-(x-\frac{3}{4})^2} = \sqrt{4+3x-2x^2}$.)

Type $\int (px+q)\sqrt{ax^2+bx+c}\,dx$ (§3.3.2)

Write $px+q = A\frac{d}{dx}(ax^2+bx+c) + B$. The A-part is $\int D'\sqrt{D}\,dx = \frac{2}{3}D^{3/2}$; the B-part is one of the three root formulas. Example: $\int (x+1)\sqrt{2x^2+3}\,dx = \frac{1}{6}(2x^2+3)^{3/2} + \sqrt{2}\left[\frac{x}{2}\sqrt{x^2+\frac{3}{2}} + \frac{3}{4}\log\left|x+\sqrt{x^2+\frac{3}{2}}\right|\right] + c$.

9.3 The $e^x[f(x)+f'(x)]$ shortcut (§3.3.3)

Result

$$\int e^x\,[f(x)+f'(x)]\,dx = e^x f(x)+c$$

Proof

$\frac{d}{dx}[e^x f(x)] = e^x f'(x)+f(x)e^x = e^x[f(x)+f'(x)]$. By the definition of integration, the result follows.

How to spot it

You see e^x multiplied by a bracket. Split the bracket into two pieces where **one is the derivative of the other**. The piece that is *not* a derivative is $f(x)$; the answer is simply $e^x f(x)$.

Example 9.9 — $\int e^{\sin^{-1}x}\left(\frac{x+\sqrt{1-x^2}}{\sqrt{1-x^2}}\right)dx$ [PYQ Feb 2025, 4 marks]

Put $\sin^{-1}x = t$: $x=\sin t$, $\frac{dx}{\sqrt{1-x^2}} = dt$, and $\sqrt{1-x^2}=\cos t$.

$$I = \int e^t(\sin t+\cos t)\,dt.$$

With $f(t)=\sin t$, $f'(t)=\cos t$:

$$I = e^t\sin t + c = x\,e^{\sin^{-1}x}+c.$$

Example 9.10 — $\int f(x)g(x)\,dx$ with $f=\frac{\sin^{-1}x}{\sqrt{1-x^2}}$, $g=e^{\sin^{-1}x}$ [PYQ Jun 2026 MCQ; also Misc. Ex 3 (I)(7)]

Put $\sin^{-1}x=t$, $\frac{dx}{\sqrt{1-x^2}}=dt$: $I=\int t\,e^t\,dt = \int e^t(t+1-1)\,dt = e^t(t-1)+c$ (or by parts).

$$I = e^{\sin^{-1}x}\left(\sin^{-1}x-1\right)+c \quad \text{— option (b).}$$

Example 9.11 — Textbook set

1. $\int e^x\frac{2+\sin 2x}{1+\cos 2x}dx = \int e^x\frac{2+2\sin x\cos x}{2\cos^2 x}dx = \int e^x(\sec^2 x+\tan x)\,dx = e^x\tan x+c$.
2. $\int e^x\frac{x+2}{(x+3)^2}dx = \int e^x\left[\frac{1}{x+3}-\frac{1}{(x+3)^2}\right]dx = \frac{e^x}{x+3}+c$.
3. $\int e^{\tan^{-1}x}\frac{1+x+x^2}{1+x^2}dx$: put $\tan^{-1}x=t$ to get $\int e^t(\tan t+\sec^2 t)\,dt = e^t\tan t = x\,e^{\tan^{-1}x}+c$.
4. $\int\frac{(x^2+1)e^x}{(x+1)^2}dx = \int e^x\left[\frac{x-1}{x+1}+\frac{2}{(x+1)^2}\right]dx = e^x\frac{x-1}{x+1}+c$ (write $x^2+1=(x^2-1)+2$).
5. $\int\left[\log(\log x)+\frac{1}{(\log x)^2}\right]dx = x\log(\log x)-\frac{x}{\log x}+c$ (textbook solved Ex. 5 — by parts on $\log(\log x)\cdot 1$, and once more on $\frac{1}{\log x}\cdot 1$; the two $\int\frac{dx}{(\log x)^2}$ terms cancel).

9.4 Generalised integration by parts (§3.5 "Something interesting")

Tabular form

$$\int u\,v\,dx = u\,v_1 - u'\,v_2 + u''\,v_3 - u'''\,v_4 + \cdots$$

where dashes are derivatives of u and subscripts are repeated integrals of v. Use it when u is a polynomial (its derivatives eventually become 0).
Example: $\int x^2\cos 3x\,dx = x^2\frac{\sin 3x}{3} - (2x)\left(-\frac{\cos 3x}{9}\right) + (2)\left(-\frac{\sin 3x}{27}\right) = \frac{1}{3}x^2\sin 3x + \frac{2}{9}x\cos 3x - \frac{2}{27}\sin 3x + c$.

Exam Tip

Use the tabular form to *check* your answer quickly, but in a 4-mark board answer show at least the first application of the formula in full (formula line + substitution), as in Example 9.3. Method marks are attached to the formula step.

10. Partial Fractions (§3.4, Ex 3.4)

Intuition — un-adding fractions

$\frac{1}{x-1}+\frac{2}{x+2}=\frac{3x}{(x-1)(x+2)}$. Integrating the *right* side is hard; integrating the *left* side is two logs. Partial fractions is the reverse of adding fractions: break a complicated rational function into simple pieces you can integrate.

Step 0 — proper or improper?

$\frac{f(x)}{g(x)}$ is **proper** if degree of f < degree of g. If not, first divide: $\frac{f}{g}=\text{Quotient}+\frac{\text{Remainder}}{g}$, then split the proper part.

The three forms (§3.4 table)

Denominator	Partial fraction form
(i) Non-repeated linear $(x-a)(x-b)(x-c)$	$\frac{A}{x-a}+\frac{B}{x-b}+\frac{C}{x-c}$
(ii) Repeated linear $(x-a)^2(x-b)$	$\frac{A}{x-a}+\frac{B}{(x-a)^2}+\frac{C}{x-b}$
(iii) Linear × irreducible quadratic $(x-a)(x^2+bx+c)$	$\frac{A}{x-a}+\frac{Bx+C}{x^2+bx+c}$

Finding constants. Multiply both sides by the full denominator, then:

- put each **root** of a linear factor (fastest), and
- **compare coefficients** of powers of x for the remaining constants (needed for repeated and quadratic factors).

Example 10.1 — $\int \frac{3x^2+4x-5}{(x^2-1)(x+2)}\,dx$ [PYQ Feb 2026, 4 marks = Textbook solved Ex. 1]

$(x^2-1)(x+2)=(x-1)(x+1)(x+2)$ — three distinct linear factors (type i).

$$\frac{3x^2+4x-5}{(x-1)(x+1)(x+2)}=\frac{A}{x-1}+\frac{B}{x+1}+\frac{C}{x+2}$$

$$\Rightarrow\ 3x^2+4x-5=A(x+1)(x+2)+B(x-1)(x+2)+C(x-1)(x+1).$$

- $x=1$: $3+4-5=A(2)(3)\Rightarrow 2=6A\Rightarrow A=\frac{1}{3}$.
- $x=-1$: $3-4-5=B(-2)(1)\Rightarrow -6=-2B\Rightarrow B=3$.
- $x=-2$: $12-8-5=C(-3)(-1)\Rightarrow -1=3C\Rightarrow C=-\frac{1}{3}$.

$$I=\int\left[\frac{1/3}{x-1}+\frac{3}{x+1}-\frac{1/3}{x+2}\right]dx=\frac{1}{3}\log|x-1|+3\log|x+1|-\frac{1}{3}\log|x+2|+c$$

$$=\frac{1}{3}\log\left|\frac{(x-1)(x+1)^9}{x+2}\right|+c.$$

Example 10.2 — Repeated factor: $\int \frac{3x-2}{(x+1)^2(x+3)}\,dx$ (Ex 3.4 I.13)

$$\frac{3x-2}{(x+1)^2(x+3)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+3} \Rightarrow 3x-2 = A(x+1)(x+3) + B(x+3) + C(x+1)^2.$$

- $x=-1$: $-5 = 2B \Rightarrow B = -\frac{5}{2}$.
- $x=-3$: $-11 = 4C \Rightarrow C = -\frac{11}{4}$.
- Coefficient of x^2: $0 = A + C \Rightarrow A = \frac{11}{4}$.

$$I = \frac{11}{4}\log|x+1| - \frac{5}{2}\int (x+1)^{-2}dx - \frac{11}{4}\log|x+3| = \frac{11}{4}\log\left|\frac{x+1}{x+3}\right| + \frac{5}{2(x+1)} + c.$$

Note: $\int \frac{dx}{(x+1)^2} = -\frac{1}{x+1}$ — a power, not a log.

Example 10.3 — Treat x^2 as one variable: $\int \frac{2x^2-3}{(x^2-5)(x^2+4)}\,dx$ (Textbook)

Only x^2 appears, so put $x^2 = m$ *only for the splitting*: $\frac{2m-3}{(m-5)(m+4)} = \frac{A}{m-5} + \frac{B}{m+4}$, $2m-3 = A(m+4) + B(m-5)$. $m=5$: $7 = 9A$, $A = \frac{7}{9}$. $m=-4$: $-11 = -9B$, $B = \frac{11}{9}$.

$$I = \frac{7}{9}\int \frac{dx}{x^2-(\sqrt{5})^2} + \frac{11}{9}\int \frac{dx}{x^2+2^2} = \frac{7}{18\sqrt{5}}\log\left|\frac{x-\sqrt{5}}{x+\sqrt{5}}\right| + \frac{11}{18}\tan^{-1}\frac{x}{2} + c.$$

Example 10.4 — Exponential reduced to type (iii): $\int \frac{5e^x}{(e^x+1)(e^{2x}+9)}\,dx$ [PYQ Mar 2024, 4 marks]

Put $e^x = t$, $e^x\,dx = dt$: $I = \int \frac{5\,dt}{(t+1)(t^2+9)}$.
$\frac{5}{(t+1)(t^2+9)} = \frac{A}{t+1} + \frac{Bt+C}{t^2+9} \Rightarrow 5 = A(t^2+9) + (Bt+C)(t+1)$.

- $t=-1$: $5 = 10A \Rightarrow A = \frac{1}{2}$.
- Coefficient of t^2: $0 = A + B \Rightarrow B = -\frac{1}{2}$.
- Constant term: $5 = 9A + C \Rightarrow C = 5 - \frac{9}{2} = \frac{1}{2}$.

$$I = \frac{1}{2}\int \frac{dt}{t+1} - \frac{1}{2}\int \frac{t\,dt}{t^2+9} + \frac{1}{2}\int \frac{dt}{t^2+9} = \frac{1}{2}\log(t+1) - \frac{1}{4}\log(t^2+9) + \frac{1}{2}\cdot\frac{1}{3}\tan^{-1}\frac{t}{3} + c$$

$$= \frac{1}{2}\log(e^x+1) - \frac{1}{4}\log(e^{2x}+9) + \frac{1}{6}\tan^{-1}\left(\frac{e^x}{3}\right) + c.$$

Example 10.5 — Trig reduced to partial fractions: $\int \frac{d\theta}{\sin\theta(3+2\cos\theta)}$ (Textbook)

Multiply top and bottom by $\sin\theta$: $\int \frac{\sin\theta\,d\theta}{(1-\cos^2\theta)(3+2\cos\theta)}$. Put $\cos\theta = t$, $\sin\theta\,d\theta = -dt$:

$$I = \int \frac{-dt}{(1-t)(1+t)(3+2t)}.$$

$-1 = A(1+t)(3+2t) + B(1-t)(3+2t) + C(1-t)(1+t)$. $t=1$: $-1 = 10A$, $A = -\frac{1}{10}$. $t=-1$: $-1 = 2B$, $B = -\frac{1}{2}$. $t=-\frac{3}{2}$: $-1 = C\left(\frac{5}{2}\right)\left(-\frac{1}{2}\right)$, $C = \frac{4}{5}$.

$$I = -\frac{1}{10}\cdot\frac{\log|1-t|}{-1} - \frac{1}{2}\log|1+t| + \frac{4}{5}\cdot\frac{\log|3+2t|}{2} + c$$

$$= \frac{1}{10}\log|1-\cos\theta| - \frac{1}{2}\log|1+\cos\theta| + \frac{2}{5}\log|3+2\cos\theta| + c.$$

Example 10.6 — Multiply to create the derivative: $\int \frac{dx}{x(x^5+1)}$ (Ex 3.4 I.8)

Multiply top and bottom by x^4: $\int \frac{x^4\,dx}{x^5(x^5+1)}$. Put $x^5 = t$, $5x^4dx = dt$:

$$I = \frac{1}{5}\int \frac{dt}{t(t+1)} = \frac{1}{5}\int \left(\frac{1}{t} - \frac{1}{t+1}\right) dt = \frac{1}{5}\log\left|\frac{x^5}{x^5+1}\right| + c.$$

Example 10.7 — Improper fraction first: $\int \frac{x^2+x-1}{x^2+x-6}\,dx$ (Ex 3.4 I.5)

Degrees are equal, so divide: $\frac{x^2+x-1}{x^2+x-6} = 1 + \frac{5}{(x+3)(x-2)}$. And $\frac{5}{(x+3)(x-2)} = \frac{1}{x-2} - \frac{1}{x+3}$.

$$I = x + \log\left|\frac{x-2}{x+3}\right| + c.$$

Common Mistakes in partial fractions

1. Using $\frac{A}{x^2+bx+c}$ (constant numerator) over a quadratic — it must be $\frac{Bx+C}{\dots}$.
2. For a repeated factor, writing only $\frac{B}{(x-a)^2}$ and missing $\frac{A}{x-a}$.
3. Forgetting to divide first when the fraction is improper.
4. $\int \frac{dx}{a-x} = -\log|a-x|$ and $\int \frac{dx}{3+2t} = \frac{1}{2}\log|3+2t|$: the coefficient of the variable divides.

11. Method Selection Guide

When you see an integral, ask these questions **in order**:

#	Question	If yes →
1	Is it directly in the standard table (after splitting into terms)?	Integrate term by term
2	Can algebra/trig identities simplify it (powers, $\tan^2$, product-to-sum, conjugate, $e^{\log}$, inverse-trig of trig)?	Simplify (Ex 3.1)
3	Is numerator = (constant) × derivative of denominator, or an inner function's derivative present?	Substitution / Corollary III
4	$\frac{1}{x^2\pm a^2}$, $\frac{1}{\sqrt{\dots}}$, or a quadratic in the denominator?	Special integrals + completing the square
5	Denominator $a\sin^2 x + b\cos^2 x + c$?	Divide by $\cos^2 x$, $t = \tan x$
6	Denominator $a\sin x + b\cos x + c$?	$t = \tan\frac{x}{2}$ (or $R\sin(x \pm \alpha)$ when $c = 0$)
7	Linear numerator over quadratic (or its root)?	$N = A\,D' + B$
8	Product of two different types ($x \cdot e^x$, $x\log x$, $x\sin x$)?	By parts (LIATE)

#	Question	If yes →
9	$e^x \times [\ldots]$ where the bracket splits as $f + f'$?	$e^x f(x)$
10	Rational function with factorisable denominator?	Partial fractions

12. Master List of Common Mistakes

1. Missing $+c$.
2. Missing the $\frac{1}{a}$ when integrating $f(ax+b)$, e.g. $\int \cos 5x\,dx = \frac{\sin 5x}{5}$, not $\sin 5x$.
3. Sign errors in $\int \sin$, $\int \operatorname{cosec}^2$, $\int \operatorname{cosec} x \cot x$, $\int \frac{dx}{a-x}$.
4. Applying the power rule to $\frac{1}{x}$ ($n = -1$ is excluded — it gives log).
5. Leaving x in the integrand after substitution, or not converting dx.
6. Forgetting to resubstitute t back in terms of x in the final answer.
7. In $\frac{1}{x^2-a^2}$ vs $\frac{1}{a^2-x^2}$, putting the log fraction upside down.
8. In completing the square, adding $\left(\frac{1}{2}\text{coef}\right)^2$ without first making the x^2 coefficient 1.
9. In by parts, choosing u against LIATE and getting a harder integral.
10. Dropping brackets: $-\int(\ldots)$ — the minus applies to *every* term inside.

13. Concept Summary and Formula Sheet

One-page formula sheet

Standard: $\int x^n = \frac{x^{n+1}}{n+1}$; $\int \frac{1}{x} = \log|x|$; $\int e^x = e^x$; $\int a^x = \frac{a^x}{\log a}$; $\int \sin = -\cos$; $\int \cos = \sin$; $\int \sec^2 = \tan$; $\int \operatorname{cosec}^2 = -\cot$; $\int \sec\tan = \sec$; $\int \operatorname{cosec}\cot = -\operatorname{cosec}$.
Trig four: $\int \tan = \log|\sec x|$; $\int \cot = \log|\sin x|$; $\int \sec = \log|\sec x + \tan x|$; $\int \operatorname{cosec} = \log|\operatorname{cosec} x - \cot x|$.
Substitution: $\int \frac{f'}{f} = \log|f|$; $\int f^n f' = \frac{f^{n+1}}{n+1}$; $\int \frac{f'}{\sqrt{f}} = 2\sqrt{f}$; $\int f(ax+b) = \frac{1}{a}g(ax+b)$.
Special seven: $\frac{1}{a}\tan^{-1}\frac{x}{a}$; $\frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|$; $\frac{1}{2a}\log\left|\frac{a+x}{a-x}\right|$; $\sin^{-1}\frac{x}{a}$; $\log|x+\sqrt{x^2-a^2}|$; $\log|x+\sqrt{x^2+a^2}|$; $\frac{1}{a}\sec^{-1}\frac{x}{a}$.
Roots by parts: $\int \sqrt{a^2-x^2} = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}$; $\int \sqrt{x^2 \pm a^2} = \frac{x}{2}\sqrt{x^2 \pm a^2} \pm \frac{a^2}{2}\log|x + \sqrt{x^2 \pm a^2}|$.
Parts: $\int uv = u\int v - \int(u'\int v)$, LIATE. **Shortcut:** $\int e^x[f + f'] = e^x f$.
Exponential × trig: $\int e^{ax}\sin bx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)$; $\int e^{ax}\cos bx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx)$.

14. Self-Assessment

Attempt all ten without looking at the notes. Answers follow. Score: 9–10 excellent; 7–8 revise weak types in File 2; below 7, re-read the relevant sections here.

1. $\displaystyle\int \frac{\cos 2x}{\sin^2 x \cos^2 x}\,dx$
2. $\displaystyle\int \frac{(\log x)^n}{x}\,dx$
3. $\displaystyle\int \frac{dx}{\sqrt{x^2+8x-20}}$
4. $\displaystyle\int \frac{dx}{4+3\cos^2 x}$
5. $\displaystyle\int \frac{3x+4}{x^2+6x+5}\,dx$

6. $\displaystyle\int x^3 \log x \, dx$
7. $\displaystyle\int e^x \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$
8. $\displaystyle\int \frac{2x}{4 - 3x - x^2} \, dx$
9. $\displaystyle\int \frac{dx}{x^3 - 1}$
10. $\displaystyle\int \cos\sqrt{x} \, dx$

Answers (with the key step)

1. $\frac{\cos^2 x - \sin^2 x}{\sin^2 x \cos^2 x} = \operatorname{cosec}^2 x - \sec^2 x$; answer $-\cot x - \tan x + c$.
2. $\log x = t$: $\frac{(\log x)^{n+1}}{n+1} + c$ $(n \neq -1)$.
3. $x^2 + 8x - 20 = (x+4)^2 - 36$: $\log\left|x + 4 + \sqrt{x^2 + 8x - 20}\right| + c$.
4. Divide by $\cos^2 x$: $\int \frac{\sec^2 x}{4\tan^2 x + 7} dx = \frac{1}{2\sqrt{7}} \tan^{-1}\left(\frac{2\tan x}{\sqrt{7}}\right) + c$.
5. $3x + 4 = \frac{3}{2}(2x + 6) - 5$: $\frac{3}{2}\log|x^2 + 6x + 5| - \frac{5}{4}\log\left|\frac{x+1}{x+5}\right| + c$.
6. $u = \log x$: $\frac{x^4}{4}\log x - \frac{x^4}{16} + c$.
7. $f = \frac{1}{x}$, $f' = -\frac{1}{x^2}$: $\frac{e^x}{x} + c$.
8. $4 - 3x - x^2 = (4 + x)(1 - x)$; $\frac{2x}{(4+x)(1-x)} = \frac{2/5}{1-x} - \frac{8/5}{4+x}$: $-\frac{2}{5}\log|1 - x| - \frac{8}{5}\log|4 + x| + c$.
9. $\frac{1}{(x-1)(x^2+x+1)} = \frac{1/3}{x-1} - \frac{\frac{1}{3}x + \frac{2}{3}}{x^2+x+1}$: $\frac{1}{3}\log|x-1| - \frac{1}{6}\log(x^2 + x + 1) - \frac{1}{\sqrt{3}}\tan^{-1}\frac{2x+1}{\sqrt{3}} + c$.
10. $\sqrt{x} = t$, $dx = 2t\,dt$: $2\int t\cos t\,dt = 2(t\sin t + \cos t) = 2\sqrt{x}\sin\sqrt{x} + 2\cos\sqrt{x} + c$.